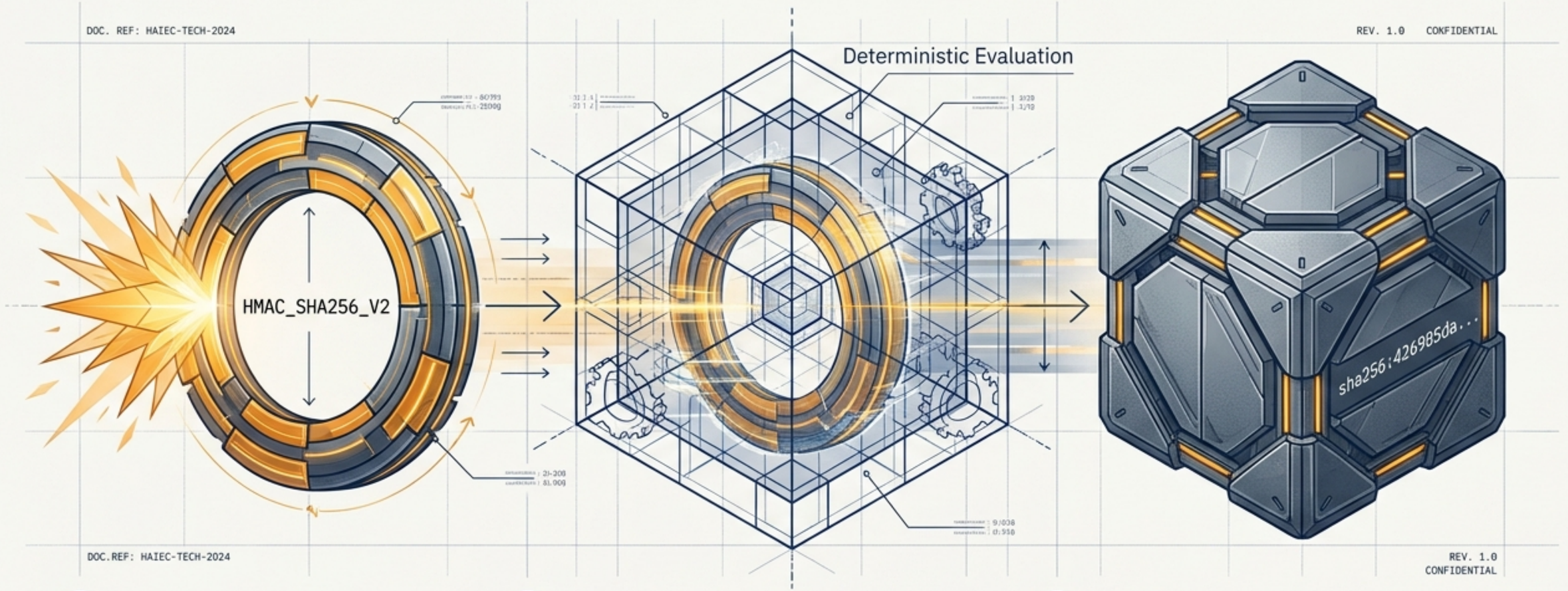
Evidence-Bound Assurance for Consequential AI

The Cryptographic Mechanics of HAIEC

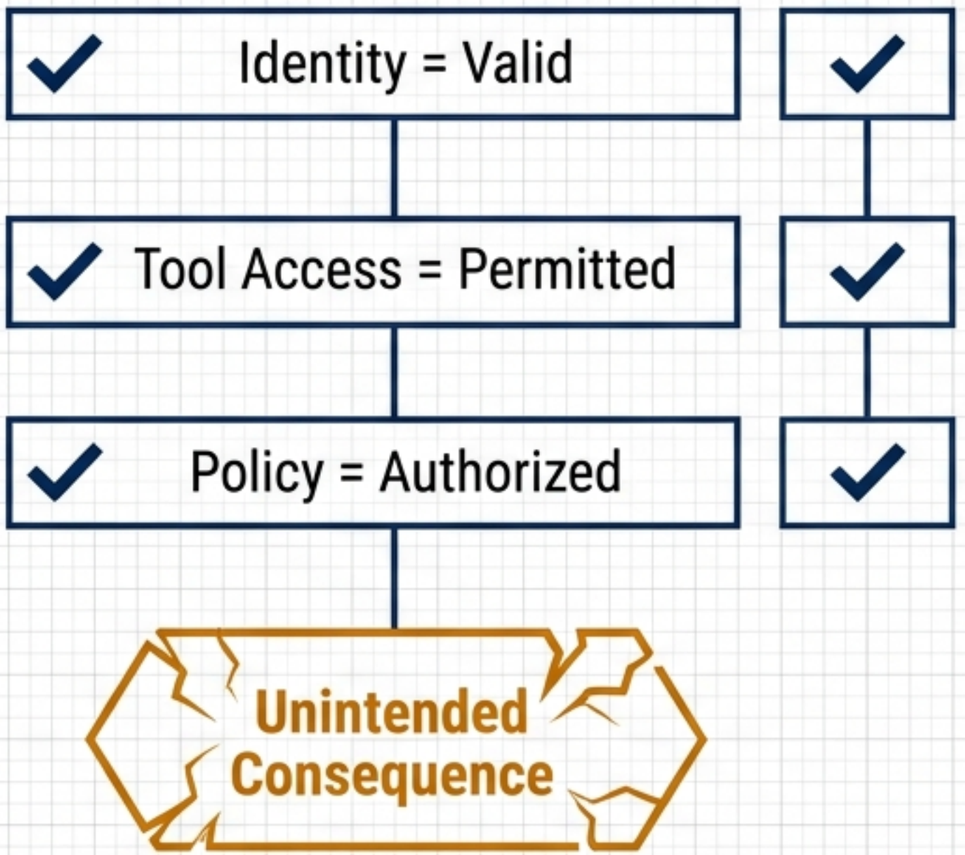


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Permission is Not Delegation

A valid tool, called by a **valid identity**, can still cause an unintended **consequence**. How do you defend the deployment decision?

Path A: The Traditional Trap



Path B: The HAIEC Defense

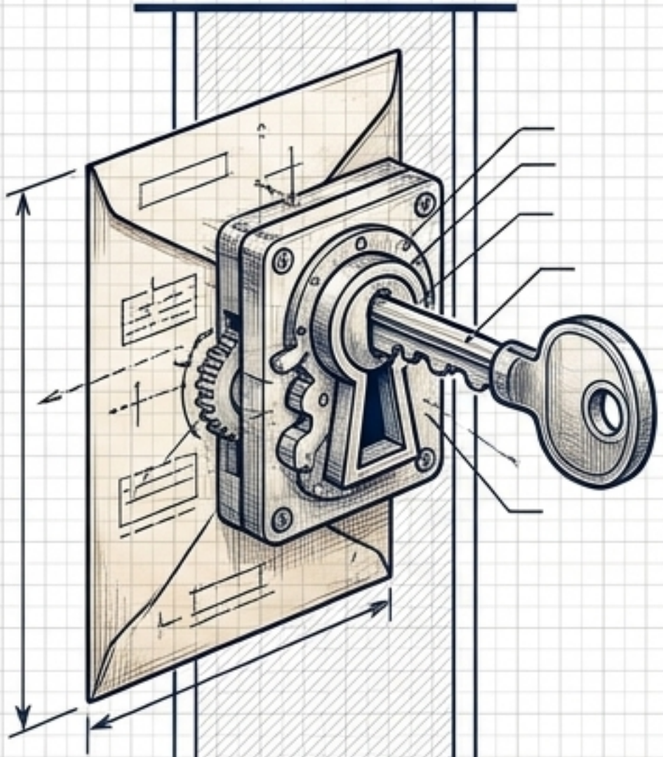


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The Architecture of Absolute Trust

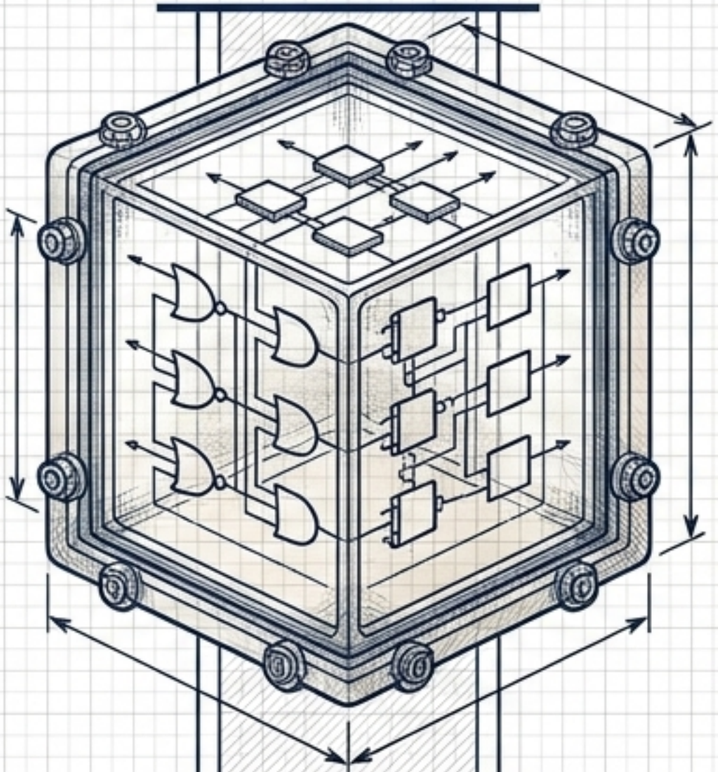
The Unassailable Chain of Custody

Input Integrity



V2 Payload-Bound Attestation

Processing Integrity



Deterministic Evaluation Engine

Output Integrity



SHA-256 Truth Digests

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Pillar 1: Input Integrity

Moving Beyond Source Identity to Consequence Locking

The Gap



Securing the origin isn't enough—we must secure the **exact consequence**.

The Solution

HMAC_SHA256_V2

- Protocol Version
- Tenant Scope
- Binder Scope
- Producer Ref
- Event ID
- payloadDigest

The **entire canonical payload** (operation, target, transaction, result) is mathe**matically bound** to the signature.

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Why V2 is Mandatory for Independent-Effect Proof

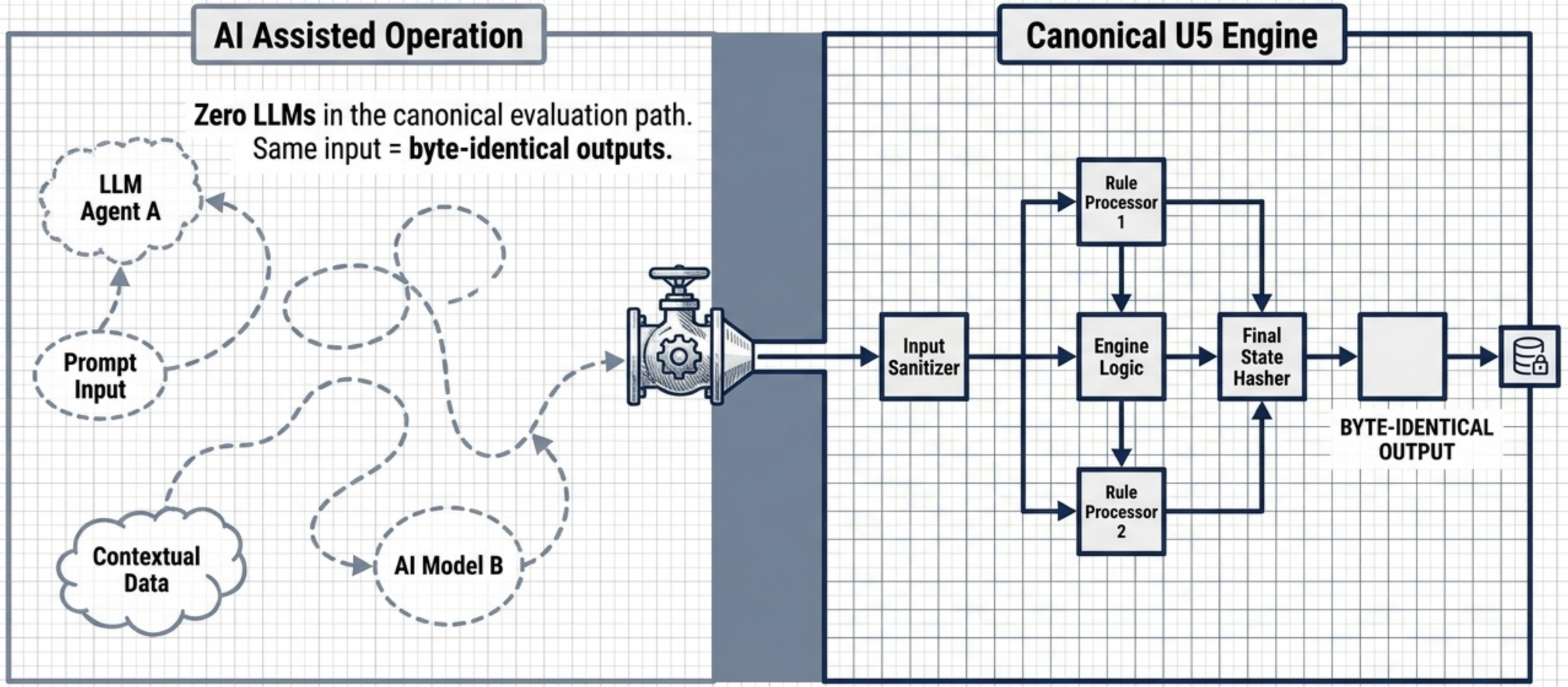
Only V2 secures the **entire canonical payload**. This makes it tamper-proof and the only protocol eligible to establish an “independent-effect witness” (UCP-08 R01 receipt).

	Legacy V1_IDENTITY	V2_PAYLOAD_BOUND
Identity Bound	Yes	**Yes**
Payload Bound	No	**Yes** (via **payloadDigest**)
Replay Resistance	**Weak** – payload is malleable	**Absolute** – tampering invalidates hash
Attestation State	**AUTHENTICATED**	**AUTHENTICATED + V2_PAYLOAD_BOUND**

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Pillar 2: Deterministic Evaluation Processing

AI Assesses. Algorithms Evaluate.



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The Five Independent Evidence Planes

Evidence planes are independent and never merged. Divergence is a factual finding, not an arbitrary verdict.

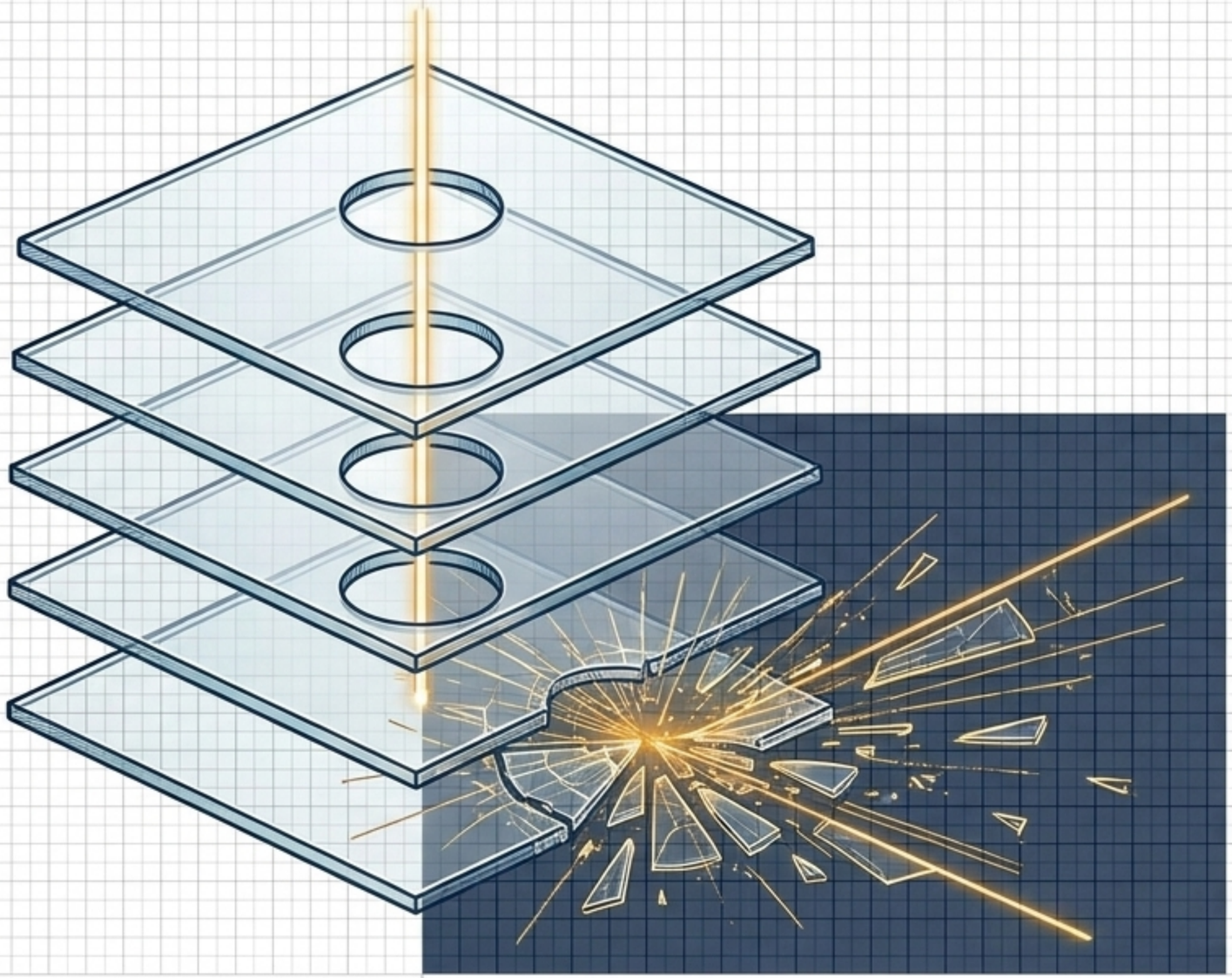
1. REQUESTED
(What the AI was asked to do)

2. POLICY-AUTHORIZED
(What rules permit)

3. EFFECTIVELY GRANTED
(What credentials allow)

4. CODE-CAPABLE
(What the application can execute)

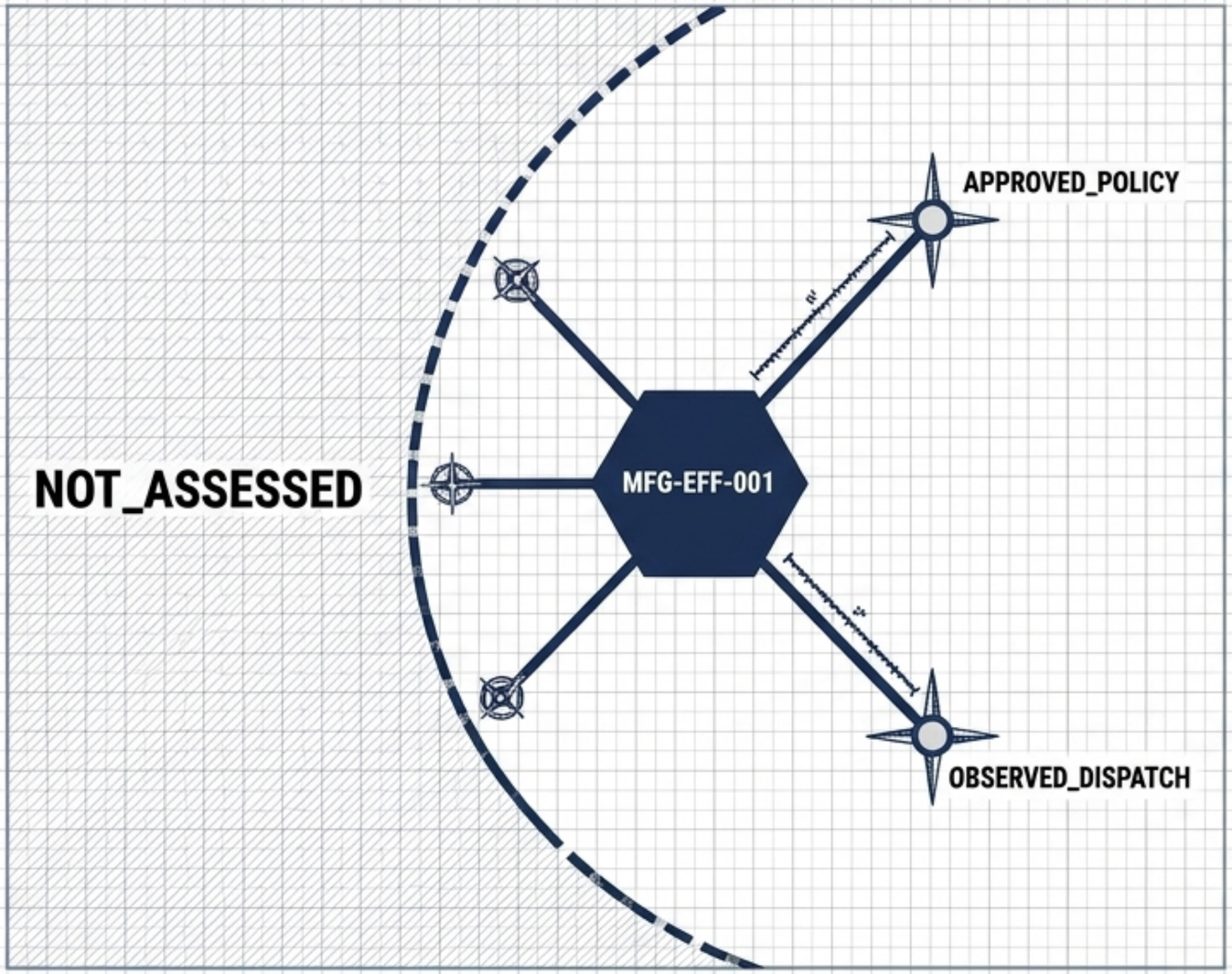
5. OBSERVED
(What actually happened at runtime)



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Rule Adjudication & The Proof Frontier

A rule **MATCH** means a factual divergence was found. If evidence is missing, the proof stops at the frontier. No silent passes.



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4 · Where does the proof stop? — PARTIAL

The proof stops at 5 unresolved evidence-frontier item(s) and 1 open evidence need(s) — listed verbatim.

- 5 evidence-frontier item(s) in state NOT_ASSESSED.
- Producer completion: COMPLETED.
- U5 reason codes: NOT_EVALUATED, REQUESTED_NOT_AUTHORIZED.
- Not essessed: AI-PROOF-002: File counts are net represented in this artifact
- Not essessed: AI-PROOF-012: Source centainer continoity not established — repository analysis cannot see build or deployment attestation
- Not assessed: AI-PROOF-014: Exact action execution identity binding not established — source defines code, not the deployed principal
- Not assessed: AI-PROOF-015: IAM authority not established by repository analysis — provider grants are esternal to source
- Not essessed: Agent Insentery: not established in this evidence basis — agenfTepology: producer uot connected to this basis
- Not assessed: How Exch Agent Operates: not established in this evidence basis — executionArchetype: producer not connected to this basis
- Nat essessed: Blast Radius / Shared Access: net established in thie evidence basis — agentTopology: producer uot connected to this basis; sharedSubstrate: producer not connected to this basis; egress: producer not connected to this basis; resourceBcope: producer not connected to this basis
- Not assessed: Human Control and Delegeted Authority: not estabiiched in this evidence basis — humanControl: producer not connected te this basis; delegatedAuthority: producer not connected to thic basis
- Nat assessed: Deferred, Persistent and Replayable Actions: not estahlished in this evidence basis — deferredExecution: producer net connected to this basis; sctionCempesltian: producer not connected to this basiy; replay: producer uet connected te thic basls
- Not assessed: Memory, Retrieval and Model-Contest Influence: not established in this evidence basis — memoryContext: producer not connected to this basis
- Not assessed: Evaluation Integrity / Self-Interest Surfaces: not established in this evidence basis — evaluationIntegrityAccess: predocer
  not connected to this basis
- Not assessed: Security Posture: not established in this evidence basis — securityScan: producer not connected to this basis; conseqeencelielts: producer net connected to this basis
- Not assessed: Sensitive Dats and Date Beondary: not established in this evidence basis — memoryContext: producer not connected to this basis
- Not assessed: Recovery, Containment and Reversibility: not established in this evidence basis — deferredExecution: producer not connected te this basis
- Not essessed: Planoer, Executor and Objective Continuity: not established in this evidence basis — delegatedAuthority: producer not connected to this basis
- Not assessed: Toel, MCP and Supply-Chain Provenance: not established in this evidence basis — capabilityChain: producer not connected te this basis
- Net assessed: Operalional Auditability and Runtime Witness: not established in this evidence basis — provenance: producer not cennected to lbis basis

Next evidence

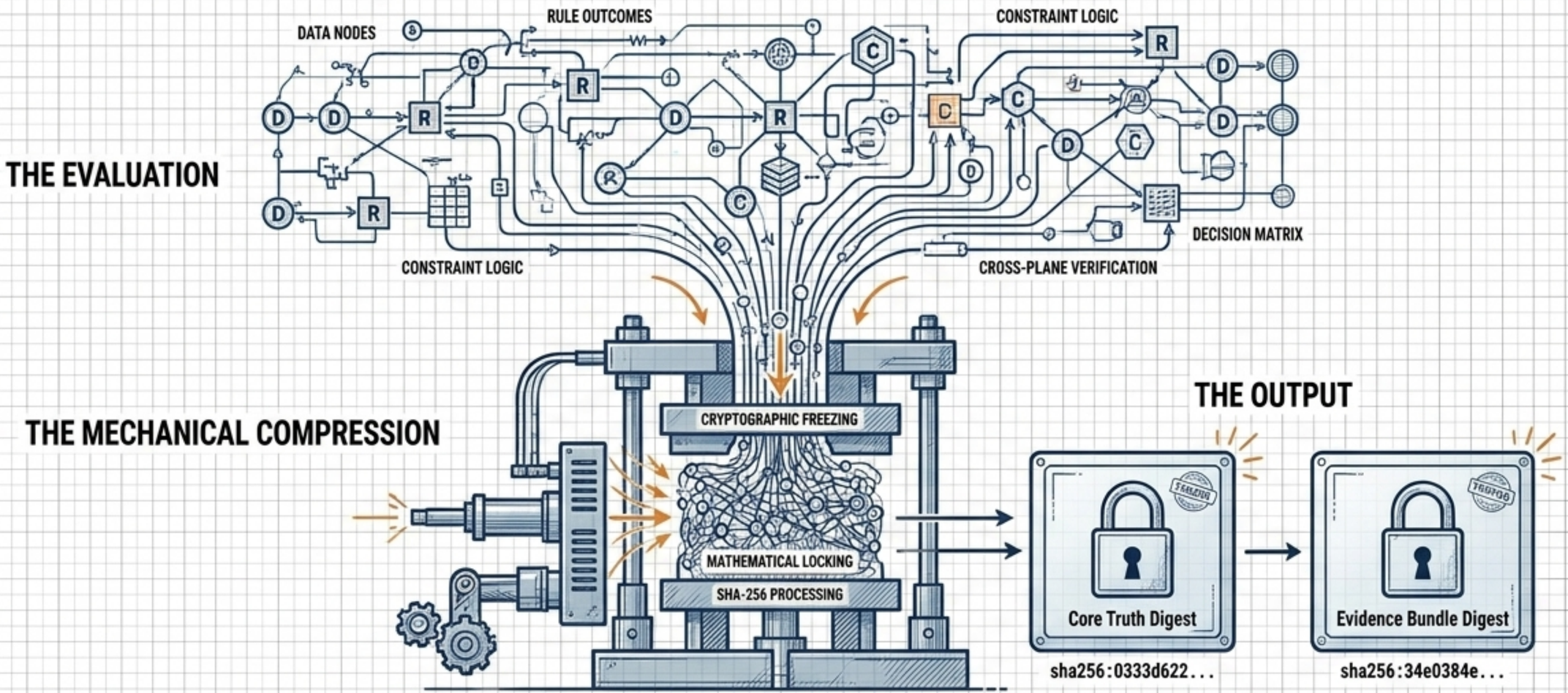
- Requested capability was not established as authorized by policy evidence.

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Pillar 3: SHA-256 Truth Digests

Integrity by Construction, Frozen in Cryptographic Amber

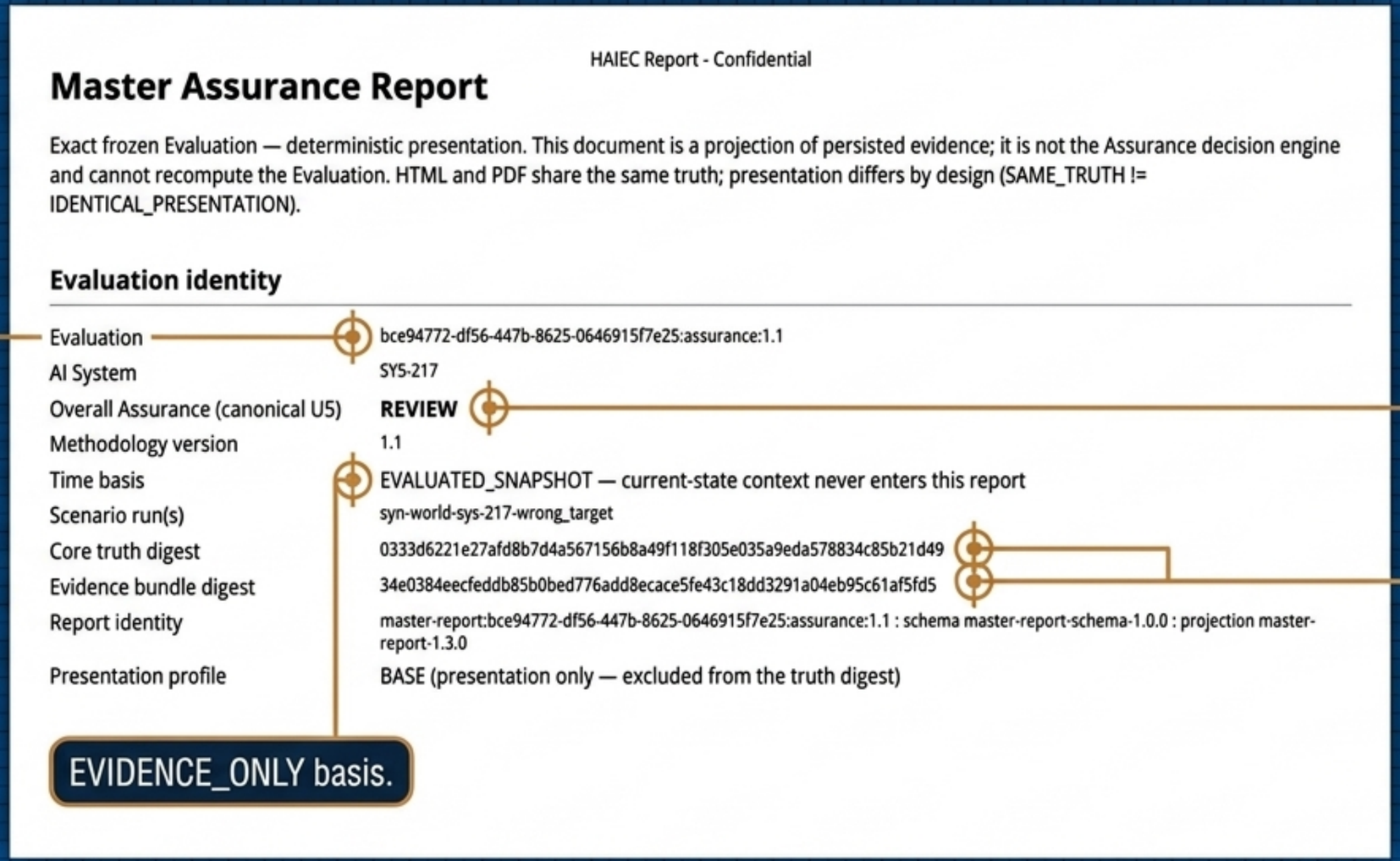
Persisted evaluation snapshots are frozen. They prove **integrity mathematically**, locking the exact state of what was evaluated.



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Anatomy of a Frozen Evaluation

This is not an assumption. It is an exact projection of persisted evidence, completely divorced from hallucination.



HAIEC Report - Confidential

Master Assurance Report

Exact frozen Evaluation — deterministic presentation. This document is a projection of persisted evidence; it is not the Assurance decision engine and cannot recompute the Evaluation. HTML and PDF share the same truth; presentation differs by design (SAME_TRUTH != IDENTICAL_PRESENTATION).

Evaluation identity

Evaluation	bce94772-df56-447b-8625-0646915f7e25:assurance:1.1
AI System	SYS-217
Overall Assurance (canonical U5)	**REVIEW**
Methodology version	1.1
Time basis	EVALUATED_SNAPSHOT — current-state context never enters this report
Scenario run(s)	syn-world-sys-217-wrong_target
Core truth digest	0333d6221e27afd8b7d4a567156b8a49f118f305e035a9eda578834c85b21d49
Evidence bundle digest	34e0384eecfeddb85b0bed776add8ecace5fe43c18dd3291a04eb95c61af5fd5
Report identity	master-report:bce94772-df56-447b-8625-0646915f7e25:assurance:1.1 : schema master-report-schema-1.0.0 : projection master-report-1.3.0
Presentation profile	BASE (presentation only — excluded from the truth digest)

Unique deterministic run.

REVIEW disposition.

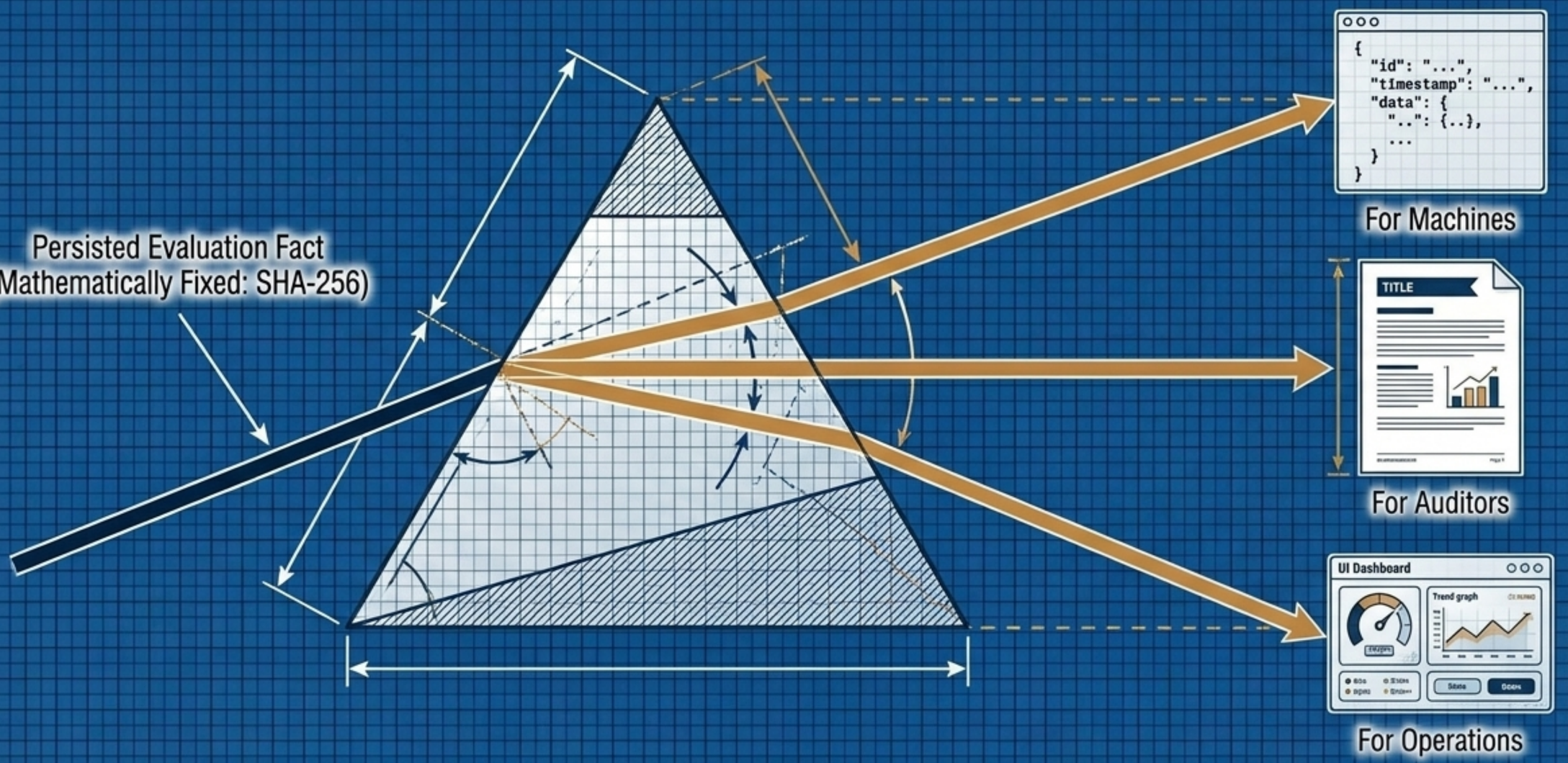
Cryptographic anchors locking the document.

EVIDENCE_ONLY basis.

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SAME_TRUTH != IDENTICAL_PRESENTATION

Presentation differs by design to serve different audiences. The underlying cryptographic truth never deviates by a single byte.



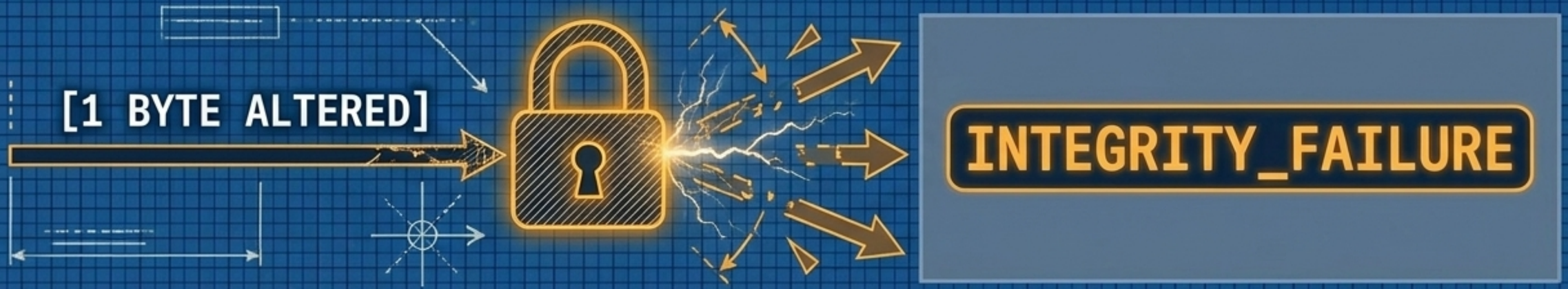
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The Fail-Closed Contract

No graceful degradation of truth. If an evaluation is tampered with, unauthorized, or missing evidence, the system fails closed. Absolute blackout.



Digest matches → Content renders flawlessly.



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‘We Govern Actions’ Is Not Enough

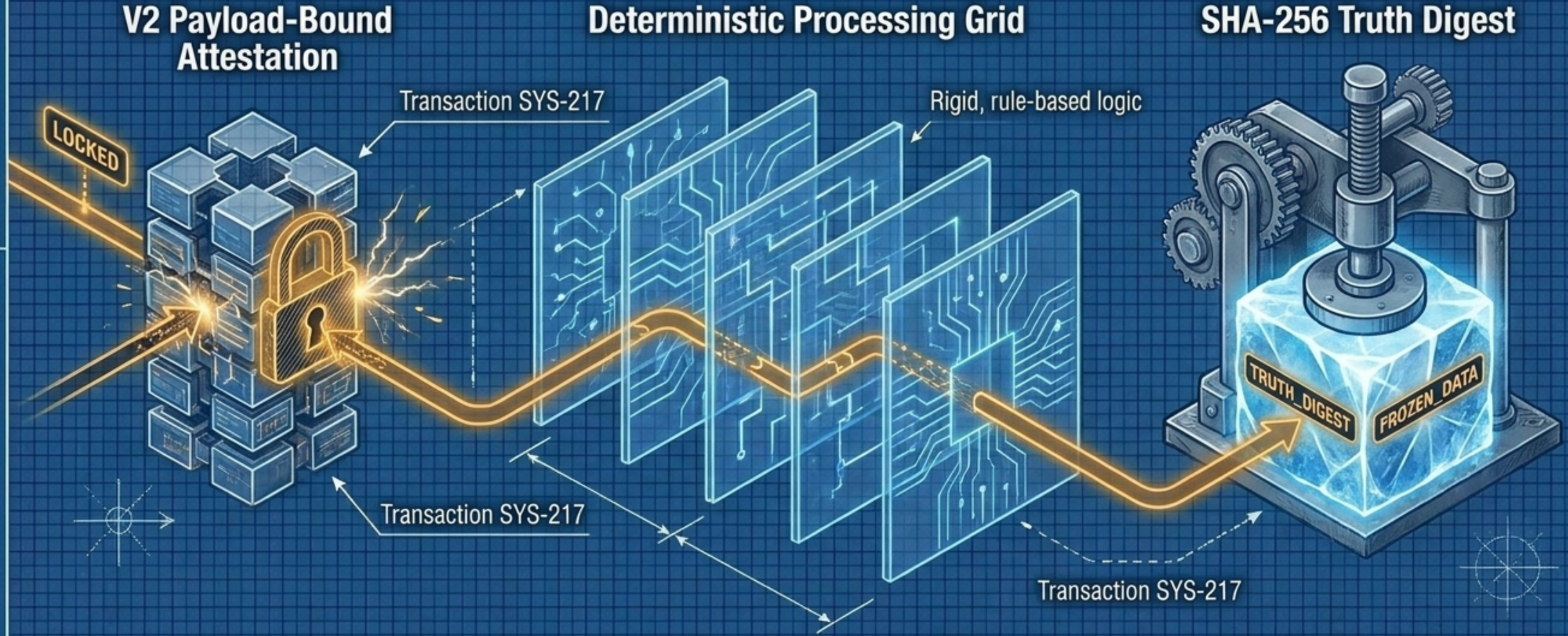
Traditional tools establish what is possible and what is permitted. Only HAIEC establishes what was actually caused, backed by independent-effect evidence.

	Scanners / Governance	IAM / Gateways	HAIEC (Evidence-Bound Assurance)
Influence Chain	Pattern matching	None	Tracks influence from prompt to event
Consequence Graph	Policy text only	Permitted access	Traces exact path to actual business consequence
Evidence Frontier	Counts vulnerabilities	Logs authentication	Maps where proof explicitly stops

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The End-to-End Bounded Path

From cryptographically signed origin to a mathematically frozen conclusion, with zero LLM hallucination.



Gemini Notebook

Defend Your Deployment

Know exactly what your AI can cause, and know exactly where the proof stops.

Scope an Enterprise Assurance POC

Evaluate one consequential AI system against a defined assurance question. Receive technical findings, inspectable evidence, explicit gaps, and a prioritized action plan.

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